TRINH DANG KHOA (ALEX TRINH)

Automotive & Embedded Systems Software Engineer

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EXPERIENCE

FPT Software (FPT Automotive Division) — **Sep 2026 – Present (Expected completion Dec 2026)**

Automotive Software Engineer Intern — *On-site, Ho Chi Minh City, Vietnam*

- Developing and verifying embedded automotive software components compliant with **AUTOSAR Classic** architecture (**BSW**, **MCAL**, **RTE**).
- Implementing and testing low-level microcontroller peripheral drivers and in-vehicle network communication stacks (**CAN**, **SPI**, **UART**) on automotive-grade MCUs (**NXP S32K144 / ARM Cortex-M**).
- Executing hardware-in-the-loop (**HIL**) bench validation, frame timing analysis, and diagnostic protocol verification.

EDUCATION

FPT University — **2024 – 2027 (Year 3 of 4)**

Bachelor of Information Technology, Major in Automotive Software Engineering (Cong nghe O to so – ASE) — *Ho Chi Minh City, Vietnam*

FPT Academy International (FAI) – Cranes Varsity — **2025 – 2027 (Year 2 of 2)**

Advanced Diploma in Automotive Software Development (Cranes Program) — *Ho Chi Minh City, Vietnam*

TECHNICAL SKILLS

Embedded & Automotive:	Embedded C, C++17, AUTOSAR Classic (BSW, MCAL, RTE), ARM Cortex-M (NXP S32K144, STM32F407, STM32F411, STM32F103), Bare-Metal & STM32 HAL, DMA, NVIC Interrupts, Timers/PWM, ADC, Watchdog
In-Vehicle Networking:	CAN 2.0A/2.0B (bxCAN, Hardware Filter Banks, Priority Arbitration), UART/USART, SPI, I2C, OBD-Style DTC Diagnostics, Hardware-in-the-Loop (HIL) Validation
HMI, Software & Tooling:	Qt 6, QML, Python, C# (.NET Native Messaging / TCP IPC), CMake, Git, STM32CubeIDE, S32 Design Studio, Keil uVision, Logic & CAN Bus Analyzers

ENGINEERING PROJECTS

Smart Vehicle Dashboard Cluster with Multi-Node CAN Bus | ***Team Lead & System Architect*** — `github.com/tumo07/vehicle-dashboard-cluster`

- Architected a distributed 3-node STM32 network (Central ECU on **STM32F407VET6** + Left/Right Body Control Modules on **STM32F411/F103**) communicating over **CAN 2.0B (500 kbps)** with a **Qt 6 / QML** digital instrument cluster and Python CAN sniffer/injector.
- Engineered priority-grouped **CAN v3.0** frame arbitration (`0x100` – `0x720`), state-machine command validation, 1 Hz node heartbeat watchdogs, OBD-style DTC fault aggregation (`B1001` – `C1011`), and a **500ms master blink tick** (`0x130`) lockstep synchronization protocol.

Portable MIFARE Classic 1K RFID Cloner & Security Research Tool — `github.com/tumo07/bluepill-rfid-cloner`

- Developed bare-metal/HAL firmware on **STM32F103C8T6 (Blue Pill)** interfacing with an **MFRC522** transceiver over **SPI** and an **SSD1306 OLED UI** over I2C.
- Implemented an automated 166-key dictionary attack, **Crypto-1** sector authentication, **Gen1a** magic backdoor (`0x40` / `0x43`) unlocking, and **Gen2/CUID** direct Block 0 UID rewrites.

LiFePO4 Battery Management & Coulomb Counting Monitor — `github.com/tumo07/stm32-battery-monitor`

- Built firmware on **STM32F103** interfacing with a **MAX17048** fuel gauge over **I2C** for **4S3P LiFePO4** battery packs, providing real-time state-of-charge (**SoC**), voltage telemetry, and fault thresholding.

Antigravity Promptless Browser Automation Bridge — `github.com/tumo07/antigravity-browser-bridge`

- Engineered a **C# Native Messaging Host** and **Manifest V3 Chrome Extension** bridging the Chrome DevTools Protocol (**CDP**) over local TCP IPC (`127.0.0.1:18789`) for autonomous agentic browser execution.

CERTIFICATIONS & LANGUAGES

IELTS Academic — Overall Band 7.0 (CEFR Level C1): Listening 8.0, Reading 8.0, Writing 6.5, Speaking 6.0 — British Council | TRF: `23VN509163TRID028A` | ielts.org/verify

CCNA: Introduction to Networks — Routing, Switching, IPv4/IPv6 Subnetting & OSI Network Architecture — Cisco

Software Development Lifecycle Specialization — Requirements, Architecture & Verification — Coursera / University of Minnesota

Languages: Vietnamese (Native), English (Professional Working Proficiency — IELTS Academic 7.0 / CEFR Level C1)